Justin "Avery" Chan

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EXPERIENCE

Amplitude
San Francisco, CA

Senior Software Engineer
May 2022 – Present

Senior Software Engineer at Amplitude (promoted from Product Engineer II in April 2026), a digital analytics platform that helps companies understand user behavior across their products to improve engagement, retention, and growth through data-driven insights. Built systems that make complex analytical systems accessible, translating complicated system configurations and computation definitions into usable, understandable interfaces across products like Data Tables, Metrics, and AI-powered chart interfaces. Worked across the full stack, primarily in React and TypeScript, building GraphQL endpoints and back-end configuration. Focused on developing analytical tools that were powerful, resilient, and accessible.

- **Feature Development on Core Products:**
  - Built and maintained **core analytics features** such as Data Tables, Metrics, and Chart Controls, enabling users to define analytical queries through intuitive UI controls reused across multiple product areas.
  - Built **pre-built analytics experiences** including Marketing and Product Analytics, and led development of the Ad Performance Hub, translating business-specific needs into guided analytical flows.
  - Developed **Out-of-the-Box Metrics** to give users default insights without setup, improving onboarding and product accessibility.
- **System Improvements:**
  - Co-created **Chart Chat**, an AI-powered interface that lets users modify charts through natural language (started as a hackathon project), later expanded into the company's **Global Chat** initiative integrating LLMs and MCP servers.
  - Implemented and maintained **system improvements** such as Dark Mode (via new GraphQL endpoints), refactored breakdown tables (tabular data that appears under all charts) to use AG Grid, and quick, high-impact UI wins like adding tooltips, multiple selection capabilities, and string search bars to improve user interaction and trust.
- **Collaboration:**
  - **Maintained team surface areas** with bug fixes, clarifying limitations, and making quick fixes for both internal and external users; supported reliability as an **on-call engineer** (debugging Cypress tests, managing deploys, coordinating rollbacks) and supported cross-org efforts including pricing and packaging.
  - Collaborated daily with product, design, and customers to come up with creative UI solutions during development and bug resolution, participating in **customer calls and bug safaris** to uncover usability issues and propose technical solutions aligned with real-world needs; onboarded engineers.
- **Full-Stack Growth:**
  - Remained a front-end specialist while growing into a full-stack contributor across design, product, and engineering, integrating **LLM, GraphQL, and back-end configuration work** into the analytics ecosystem, valuing the ability to contribute at all levels, stay unblocked, and understand the complete system.

Halo Science
Chicago, IL (remote)

Developer Intern
May 2021 – August 2021

Halo is a startup focused on pairing researchers to various funding sources. Implemented features and designs at the direction of the CEO or UX designer, fixed bugs, and set up developer productivity tools like Storybook and Chromatic. Developed a new React component library to reduce code redundancy and wrote Python web scraping scripts for the marketing team.

MAterials Simulation Toolkit Machine Learning (MAST-ML)
Madison, WI

Member
June 2020 – December 2021

MAST-ML is an open-source machine learning Python package with a focus on the material sciences. Implemented various functionality proposed by supervisor Dr. Ryan Jacobs, including data-twin removal to improve model performance, graphs to summarize classification models, and exploration of gp-learn library and other feature generation methods.

EDUCATION

University of Wisconsin-Madison Madison, WI

B.S. Major in Computer Science | B.S. Major in Data Science *August 2019 – May 2022*

Major GPA: 3.95 | Cumulative GPA: 3.75 | Dean's List 3 semesters

PROJECTS AND SKILLS

Projects selected to show a variety of project topics and programming languages. Check out more on my website or GitHub.

- **Personal Website:** Set up my website using Jekyll and wrote Python scripts to scrape GitHub (GitHub).
- **Work Visualization:** Visualized a year of self-reported data on where I spent my time working, exploring personal informatics (GitHub).
- **Antibiotic Visualization:** Made a static visualization that effectively communicates antibiotics data - one of many data visualization projects from a favorite college class on data visualization fundamentals (GitHub).
- **Python Competency:** Extensive experience with Python for automation scripts, data analysis projects, API integrations, and command-line tools - visit my website or GitHub for examples including todo migration scripts, calendar generation, and data analysis projects.
- **Scripting:** Enjoy writing Python, HTML, and JavaScript scripts for quick automation and problem-solving, excited that modern AI tools have made this both more accessible, rapid, and powerful.